

SOC ANALYSIS CAPSTONE PROJECT

Project Title

**SOC THREAT DETECTION IN A SEGMENTED HEALTH CARE
NETWORK**

Prepared By

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1. Executive Summary

MedSecure Health Solutions, a healthcare technology provider managing patient records and telemedicine services, noticed unusual activity targeting its EHR server, showing lapses in monitoring and internal controls. To strengthen security, the network was segmented into WAN, LAN, and DMZ zones, and tools such as firewalls, centralized logging, and traffic-analysis systems were deployed for improved visibility. Simulated attacks, including repeated login failures and internal scanning, were conducted to test detection and response capabilities. The assessment showed that alerts were generated quickly, visibility improved, and potential threats could be investigated more effectively. High-level recommendations include refining access controls between network segments, expanding monitoring coverage, and implementing automated response actions to enhance overall security posture.

2. Project Introduction

MedSecure Health Solutions provides healthcare technology services and manages sensitive patient data, including digital medical records and telemedicine information. Because of the critical nature of this data, the company relies on stable, secure systems to support daily operations.

Recently, unusual activity targeting the EHR server in the DMZ raised concerns about weak visibility and gaps in internal controls. These events showed that the existing setup was not fully capable of detecting or understanding potential threats.

The SOC capstone was created to assess how a structured monitoring environment could improve detection, analysis, and response. By simulating common attack behaviors and observing how the tools react, the project aims to show how segmentation, centralized logging, and alerting can strengthen the organization's overall security posture.

3. Tools Used

1. Draw.io
2. Virtualbox

3. Kali Linux
4. Pfsense
5. Wazuh (SIEM)
6. Ubuntu
7. Nmap
8. Hydra

4. METHODOLOGY

The network was first designed and segmented into three zones WAN, LAN, and DMZ to reduce exposure and better control how traffic moves between systems. A diagram of the layout was created in Draw.io to clearly show how each zone connects and which systems are allowed to communicate.

pfSense was then set up as the firewall to manage traffic between these zones. Rules were created to block unnecessary ports, restrict access, and ensure only approved communication paths were allowed. This helped create stronger boundaries and improved overall network control.

Next, the Wazuh Manager was installed on the LAN, and Wazuh agents were added to servers and endpoints. These agents collected logs such as login activity, system events, and network behavior, sending all data to the manager for centralized analysis.

To test the monitoring setup, Kali Linux was used to simulate common attacks. These tests included repeated login attempts, network scanning, and trying to gain higher access levels. This allowed the team to check whether suspicious behavior would be detected correctly.

Wireshark was also used to capture network traffic at important points in the LAN and DMZ. These captures helped identify unusual connections or unexpected activity during the simulations.

Finally, alerts from Wazuh were reviewed and compared with the network traffic to confirm whether incidents were real. Automated response features, such as blocking suspicious IP addresses, were also tested to see how quickly the system could react to potential threats.

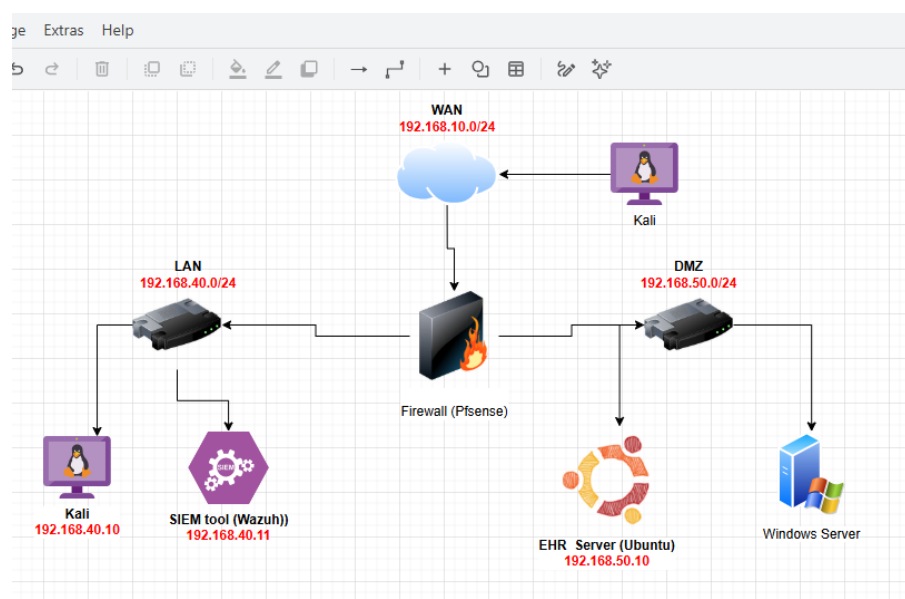
5. Phase-by-Phase Analysis

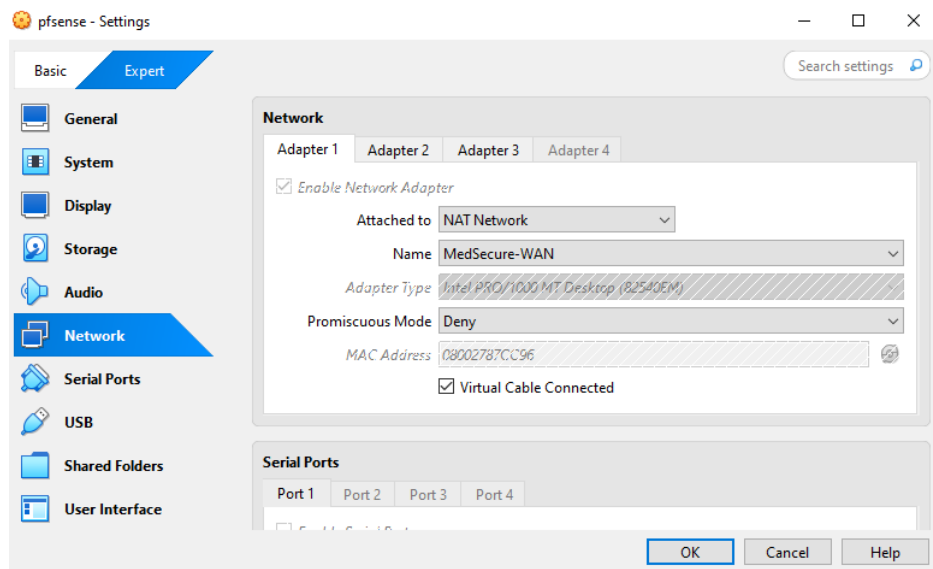
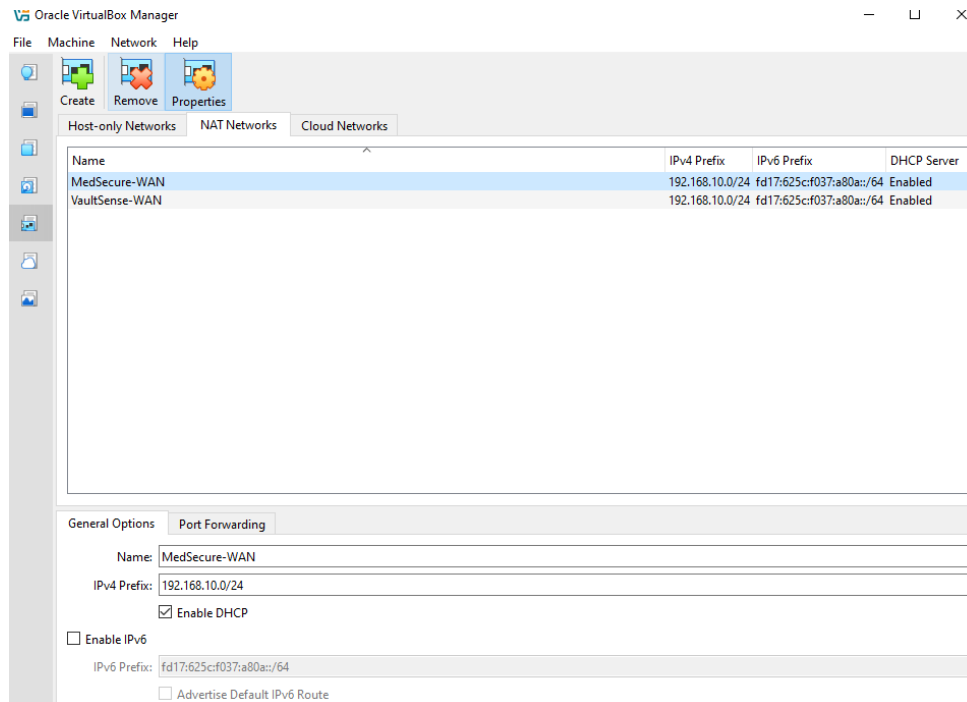
Phase 1: Network Design & Segmentation (Draw.io & pfSense)

A network diagram was created in Draw.io showing WAN, LAN, DMZ, key hosts, and the firewall to visualize traffic flows and boundaries.

In pfSense, the network was segmented to enforce clear boundaries between the zones. The WAN network, named **MedSecure**, was configured as the main source of internet connectivity, with DHCP enabled to automatically assign IP addresses. WAN, LAN, and DMZ interfaces were assigned to separate adapters (WAN: Adapter 1, LAN: Adapter 2, DMZ: Adapter 3), with DHCP enabled on the WAN.

Firewall rules were implemented to allow the LAN to access the DMZ and internet while isolating the DMZ from all other traffic. Only approved services from the WAN were allowed to the DMZ, limiting exposure and improving overall network security.





Search settings

Network

Adapter 1Adapter 2Adapter 3Adapter 4

☒ Enable Network Adapter

Attached toInternal Network

NameLAN

Adapter TypeIntel PRO/1000 MT Desktop (82540EM)

Promiscuous ModeDeny

MAC Address08002739EE70

☒ Virtual Cable Connected

Serial Ports

Port 1Port 2Port 3Port 4

OK

Cancel

Help

Search settings

network

Adapter 1Adapter 2Adapter 3Adapter 4

☒ Enable Network Adapter

Attached toInternal Network

NameDMZ

Adapter TypeIntel PRO/1000 MT Desktop (82540EM)

Promiscuous ModeDeny

MAC Address080027384C64

☒ Virtual Cable Connected

Serial Ports

Port 1Port 2Port 3Port 4

OK

Cancel

Help

```
pfSense [Running] - Oracle VirtualBox
File Machine View Input Devices Help
*** Welcome to pfSense 2.8.1-RELEASE (amd64) on MedSecure ***

WAN (wan) -> em0 -> v4/DHCP4: 192.168.10.4/24
LAN (lan) -> em1 -> v4: 192.168.40.1/24
DMZ (opt1) -> em2 -> v4: 192.168.50.1/24

0) Logout / Disconnect SSH          9) pfTop
1) Assign Interfaces                10) Filter Logs
2) Set interface(s) IP address     11) Restart GUI
3) Reset admin account and password 12) PHP shell + pfSense tools
4) Reset to factory defaults        13) Update from console
5) Reboot system                    14) Enable Secure Shell (sshd)
6) Halt system                      15) Restore recent configuration
7) Ping host                        16) Restart PHP-FPM
8) Shell

Enter an option:
Message from syslogd@MedSecure at Dec 10 00:14:21 ...
php-fpm[4351]: /index.php: Successful login for user 'admin' from: 192.168.40.11 (Local Database)

Message from syslogd@MedSecure at Dec 10 01:26:32 ...
php-fpm[4341]: /interfaces_assign.php: Successful login for user 'admin' from: 192.168.40.11 (Local Database)
```

WAN		LAN ADDRESS		PORT		ACTION		ACTION		ACTION	
				80							
☐	0/504 B	IPv4 *	LAN subnets	*	DMZ subnets	*	*	none			
☐	2/320.66 MIB	IPv4 *	LAN subnets	*	*	*	*	none	Default allow LAN to any rule		
☐	0/0 B	IPv6 *	LAN subnets	*	*	*	*	none	Default allow LAN IPv6 to any rule		

Add Add Delete Toggle Copy Save Separator

Security Controls:

- Network Segmentation
- Firewall Enforcement
- DMZ Isolation

Phase 2: Attack Simulation & Traffic Capture (Kali Linux & Wireshark)

In this phase, Kali Linux was used to simulate common attacker behaviors, including credential-guessing attempts and network scanning. User and password lists were created for testing, and a brute-force login attempt was launched against the target system to observe how the network responded. Additionally, Nmap was used to identify open services and verify the security posture of the target machine.

Wireshark was run on selected network interfaces to capture traffic during these simulated attacks. This allowed monitoring of SSH authentication attempts, scanning activity, and other suspicious packets generated by the testing processes. Important packets were flagged and exported for later correlation with the Wazuh detection logs.

```
Session Actions Edit View Help
(kali@kali)-[~]
└─$ nano users.txt
(kali@kali)-[~]
└─$ cat users.txt
admin
comfort
10alytics
medsecure
nacohort
eucohort
cybersecurity
packet
bruteforce
ubuntu
(kali@kali)-[~]
└─$ nano password.txt
(kali@kali)-[~]
└─$ cat password.txt
password12
10alytics
password
ephraim
salem
mia
canada
unitedkingdon
(kali@kali)-[~]
└─$ hydra -L users.txt -P password.txt -t4 -v ssh://192.168.50.10
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this
```

```
(kali@kali)-[~]
└─$ hydra -L users.txt -P password.txt -t4 -v ssh://192.168.50.10
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this
is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-12-09 10:54:36
[DATA] max 4 tasks per 1 server, overall 4 tasks, 132 login tries (l:1/p:12), ~33 tries per task
[DATA] attacking ssh://192.168.50.10:22/
[VERBOSE] Resolving addresses ... [VERBOSE] resolving done
[INFO] Testing if password authentication is supported by ssh://medsecureadmin@192.168.50.10:22
[INFO] Successful, password authentication is supported by ssh://192.168.50.10:22
[22][ssh] host: 192.168.50.10 login: medsecureadmin password: admin
[22][ssh] host: 192.168.50.10 login: medsecure password: medsecure
[STATUS] 88.00 tries/min, 88 tries in 00:01h, 44 to do in 00:01h, 4 active
[22][ssh] host: 192.168.50.10 login: ubuntu password: ubuntu
[STATUS] attack finished for 192.168.50.10 (waiting for children to complete tests)
1 of 1 target successfully completed, 3 valid passwords found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2025-12-09 10:56:06
```

```
(kali@kali)-[~]
└─$ sudo systemctl start ssh
[sudo] password for kali:
(kali@kali)-[~]
└─$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: disabled)
   Active: active (running) since Tue 2025-12-09 08:53:11 EST; 8s ago
     Invocation: d6267e43a7f1489081fa14264cff4201
       Docs: man:sshd(8)
             man:sshd_config(5)
    Process: 32279 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 32290 (sshd)
      Tasks: 1 (limit: 2118)
     Memory: 5.1M (peak: 5.6M)
        CPU: 52ms
     CGroup: /system.slice/ssh.service
             └─32290 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Dec 09 08:53:10 kali systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Dec 09 08:53:11 kali sshd[32290]: Server listening on 0.0.0.0 port 22.
Dec 09 08:53:11 kali sshd[32290]: Server listening on :: port 22.
Dec 09 08:53:11 kali systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
```

```
(kali@kali)-[~]
└─$ nmap -h
Nmap 7.95 ( https://nmap.org )
Usage: nmap [Scan Type(s)] [Options] {target specification}
TARGET SPECIFICATION:
  Can pass hostnames, IP addresses, networks, etc.
  Ex: scanme.nmap.org, microsoft.com/24, 192.168.0.1; 10.0.0-255.1-254
  -iL <inputfilename>: Input from list of hosts/networks
  -iR <num hosts>: Choose random targets
  --exclude <host1[,host2][,host3],...>: Exclude hosts/networks
```

```

(kali㉿kali)-[~]
$ nmap 192.168.40.0/24
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-09 09:09 EST
Nmap scan report for MedSecure.home.arpa (192.168.40.1)
Host is up (0.00096s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
53/tcp    open  domain
80/tcp    open  http
443/tcp   open  https
MAC Address: 08:00:27:39:EE:70 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap scan report for 192.168.40.10
Host is up (0.00059s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
443/tcp   open  https
MAC Address: 08:00:27:32:25:68 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap scan report for 192.168.40.11
Host is up (0.00012s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh

Nmap done: 256 IP addresses (3 hosts up) scanned in 7.72 seconds

(kali㉿kali)-[~]
$ nmap -sS 192.168.40.1
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-09 09:13 EST
Nmap scan report for MedSecure.home.arpa (192.168.40.1)
Host is up (0.00047s latency).

```

```

Nmap scan report for MedSecure.home.arpa (192.168.40.1)
Host is up (0.00047s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
53/tcp    open  domain
80/tcp    open  http
443/tcp   open  https
MAC Address: 08:00:27:39:EE:70 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 5.35 seconds

```

```

(kali㉿kali)-[~]
$ nmap -sV 192.168.40.1
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-09 09:14 EST
Nmap scan report for MedSecure.home.arpa (192.168.40.1)
Host is up (0.00057s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
53/tcp    open  domain  Unbound
80/tcp    open  http    nginx
443/tcp   open  ssl/http nginx
MAC Address: 08:00:27:39:EE:70 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 17.95 seconds

```

```

(kali㉿kali)-[~]
$ nmap -sU 192.168.40.1
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-09 09:14 EST
Nmap scan report for MedSecure.home.arpa (192.168.40.1)
Host is up (0.00055s latency).
Not shown: 997 open/filtered udp ports (no-response)
PORT      STATE SERVICE
53/udp    open  domain
123/udp   open  ntp

```

```

(kali@kali)-[~]
$ nmap -sU 192.168.40.1
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-09 09:14 EST
Nmap scan report for MedSecure.home.arpa (192.168.40.1)
Host is up (0.00055s latency).
Not shown: 997 open|filtered udp ports (no-response)
PORT      STATE SERVICE
53/udp    open  domain
123/udp   open  ntp
161/udp   open  snmp
MAC Address: 08:00:27:39:EE:70 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 13.73 seconds

(kali@kali)-[~]
$ nmap -sU -p 161 --script=snmp.interfaces 192.168.40.1
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-09 09:26 EST
Nmap scan report for MedSecure.home.arpa (192.168.40.1)
Host is up (0.00050s latency).

PORT      STATE SERVICE
161/udp   open  snmp
| snmp.interfaces:
|   em0
|     IP address: 192.168.10.4  Netmask: 255.255.255.0
|     MAC address: 08:00:27:87:cc:96 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
|     Type: ethernetCsmacd  Speed: 1 Gbps
|     Status: up
|     Traffic stats: 899.64 Kb sent, 9.64 Mb received
|   em1
|     IP address: 192.168.40.1  Netmask: 255.255.255.0
|     MAC address: 08:00:27:39:ee:70 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
|     Type: ethernetCsmacd  Speed: 1 Gbps

```

```

|     MAC address: 08:00:27:39:ee:70 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
|     Type: ethernetCsmacd  Speed: 1 Gbps
|     Status: up
|     Traffic stats: 1.85 Mb sent, 991.21 Kb received
|   em2
|     IP address: 192.168.50.1  Netmask: 255.255.255.0
|     MAC address: 08:00:27:58:4c:64 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
|     Type: ethernetCsmacd  Speed: 1 Gbps
|     Status: up
|     Traffic stats: 7.49 Mb sent, 778.03 Kb received
|   enc0
|     Type: wwanPP2  Speed: 0 Kbps
|     Status: down
|     Traffic stats: 0.00 Kb sent, 0.00 Kb received
|   lo0
|     IP address: 127.0.0.1  Netmask: 255.0.0.0
|     Type: softwareLoopback  Speed: 0 Kbps
|     Status: up
|     Traffic stats: 5.28 Kb sent, 5.28 Kb received
|   pflog0
|     Type: ifPwType  Speed: 0 Kbps
|     Status: down
|     Traffic stats: 0.00 Kb sent, 0.00 Kb received
|   pfsync0
|     Type: ilan  Speed: 0 Kbps
|     Status: down
|     Traffic stats: 0.00 Kb sent, 0.00 Kb received
|_
MAC Address: 08:00:27:39:EE:70 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.23 seconds

(kali@kali)-[~]
$ nmap 192.168.50.0/24
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-09 09:28 EST
Nmap scan report for 192.168.50.1

```

```

(kali@kali)-[~]
$ nmap 192.168.50.0/24
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-09 09:28 EST
Nmap scan report for 192.168.50.1
Host is up (0.00055s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
53/tcp    open  domain
80/tcp    open  http
443/tcp   open  https

Nmap scan report for 192.168.50.10
Host is up (0.0016s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh

Nmap done: 256 IP addresses (2 hosts up) scanned in 9.65 seconds

(kali@kali)-[~]
$ nmap 192.168.50.10
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-09 09:31 EST
Nmap scan report for 192.168.50.10
Host is up (0.0021s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh

Nmap done: 1 IP address (1 host up) scanned in 0.28 seconds

(kali@kali)-[~]
$ hydra --help
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organs
is non-binding, these ** ignore laws and ethics anyway).

```

Apply a display filter ... <Ctrl-/>							
No.	Time	Source	Destination	Protocol	Length	Info	
1	0.000000000	192.168.50.10	192.168.40.11	SSH	118	Server:	
2	0.000196838	192.168.40.11	192.168.50.10	SSH	118	Client:	
3	0.000223610	192.168.40.11	192.168.50.10	TCP	66	38850 →	
4	0.000484363	192.168.40.11	192.168.50.10	TCP	74	49122 →	
5	0.002691847	192.168.50.10	192.168.40.11	TCP	66	22 → 38	
6	0.002692163	192.168.50.10	192.168.40.11	TCP	74	22 → 49	
7	0.002744847	192.168.40.11	192.168.50.10	TCP	66	49122 →	
8	0.002800773	192.168.40.11	192.168.50.10	SSHv2	89	Client:	
9	0.006356746	192.168.50.10	192.168.40.11	TCP	66	22 → 38	
10	0.006357019	192.168.50.10	192.168.40.11	TCP	66	22 → 49	
11	0.006394322	192.168.40.11	192.168.50.10	TCP	66	38850 →	

▶ Frame 2: 118 bytes on wire (944 bits), 118

▶ Ethernet II, Src: PCSSystemtec_1f:b7:23 (08

▶ Internet Protocol Version 4, Src: 192.168.4

▶ Transmission Control Protocol, Src Port: 38

▶ SSH Protocol

0000	08 00 27 39 ee 70 08 00	27 1f b7 23 0
0010	00 68 4c 01 40 00 40 06	13 29 c0 a8 2
0020	32 0a 97 c2 00 16 80 0a	d4 81 f4 75 3
0030	01 02 db c0 00 00 01 01	08 0a aa a8 d
0040	04 e6 94 91 35 29 fb 97	9d fe 3b 6c 2
0050	a2 48 e4 20 f8 49 9b de	3b d0 b3 d4 b
0060	6c 8b bf b4 40 2c 3f be	02 f4 0f 92 f
0070	98 eb d7 dd f4 ff	

wireshark eth0O4Q1G3.pcapng

Packets: 24196 · Dropped: 0 (0.0%)

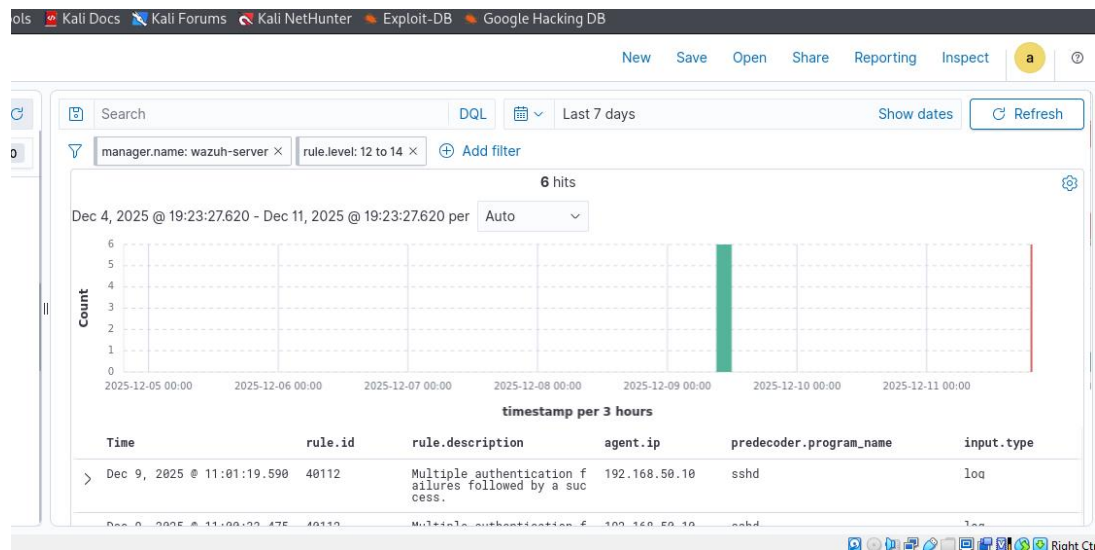
Profile: Default

wireshark_eth004Q1G3.pcapng

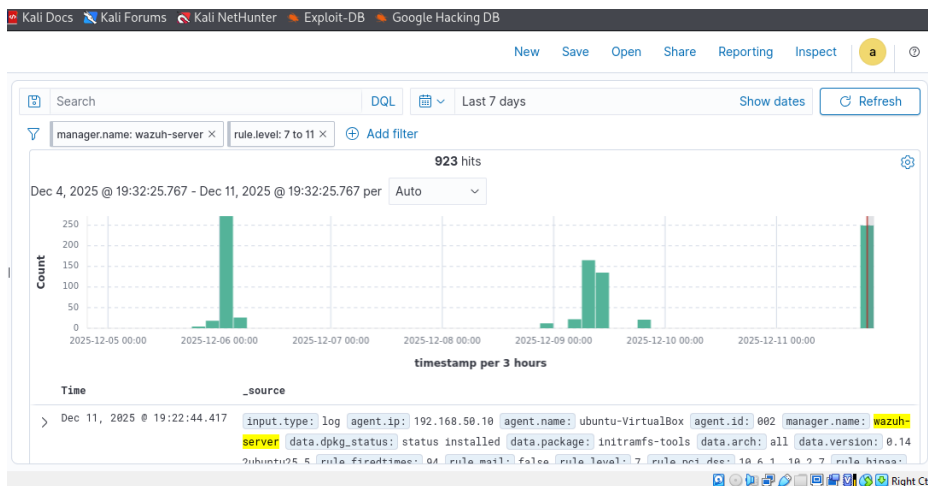
Packets: 24196 · Dropped: 0 (0.0%) · Profile: Default

Phase 3 - Security Event Monitoring & Response (Wazuh SIEM)

During Phase 3, we tested how well the security monitoring system could detect and respond to potential attacks in the environment. Wazuh collected logs from the servers and watched for unusual behaviors such as login failures, scanning activity, and repeated password attempts. When we simulated a brute-force attack using Hydra by trying many usernames and passwords on the SSH service. Wazuh immediately flagged the activity and generated high-level alerts showing multiple failed logins coming from the attacker's IP address (192.168.50.10). The alert graphs also showed clear spikes during the attack, proving that Wazuh was actively monitoring and correlating events. As the attack continued, Wazuh's incidence response feature activated and blocked the attacker's IP using a firewall rule. This was confirmed when the Hydra attack tool could no longer connect and even simple pings to the server returned 100% packet loss, meaning the attacker was completely locked out. Overall, Phase 3 demonstrated that the organization now has much better visibility into suspicious activity, fast alerting when something looks wrong, and an effective automated defense that can stop attacks before they escalate.



Time	rule.id	rule.description	agent.ip	predecoder.program_name	input.type
> Dec 9, 2025 @ 11:01:19.590	40112	Multiple authentication failures followed by a success.	192.168.50.10	sshd	log
> Dec 9, 2025 @ 11:00:33.475	40112	Multiple authentication failures followed by a success.	192.168.50.10	sshd	log
> Dec 9, 2025 @ 11:00:11.352	40112	Multiple authentication failures followed by a success.	192.168.50.10	sshd	log
> Dec 9, 2025 @ 10:57:39.387	40112	Multiple authentication failures followed by a success.	192.168.50.10	sshd	log
> Dec 9, 2025 @ 10:56:40.200	40112	Multiple authentication failures followed by a success.	192.168.50.10	sshd	log



< Manager configuration

Edit ossec.conf of Manager

```

228 <command>
229   <name>netsh</name>
230   <executable>netsh.exe</executable>
231   <timeout_allowed>yes</timeout_allowed>
232 </command>
233
234 <active-response>
235   <disabled>no</disabled>
236   <command>firewall-drop</command>
237   <location>local</location>
238   <rules_group>sshd</rules_group>
239   <timeout>0</timeout>
240 </active-response>

```

```
(kali@kali)-[~]
$ hydra -L users.txt -P password.txt -t4 -v ssh://192.168.50.10
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret ser
is non-binding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-12-09 20:22:14
[DATA] max 4 tasks per 1 server, overall 4 tasks, 132 login tries (l:11/p:12), ~33 tries per task
[DATA] attacking ssh://192.168.50.10:22/
[VERBOSE] Resolving addresses ... [VERBOSE] resolving done
[INFO] Testing if password authentication is supported by ssh://medsecureadmin@192.168.50.10:22
[ERROR] could not connect to ssh://192.168.50.10:22 - Timeout connecting to 192.168.50.10

(kali@kali)-[~]
$ ping 192.168.50.10
PING 192.168.50.10 (192.168.50.10) 56(84) bytes of data.
^C
— 192.168.50.10 ping statistics —
6 packets transmitted, 0 received, 100% packet loss, time 5185ms
```

6. FINAL FINDINGS & IMPACT

The investigation found some security weaknesses, like weak passwords and users having more access than needed. Alerts showed that SSH brute-force attempts, internal port scanning, and suspicious file-access behavior were connected to possible attacks. These issues could put sensitive data at risk, mess with system operations, and create compliance issues. Signs of a real threat included repeated login failures from certain IP addresses, users trying to get higher privileges, and unusual scans on DMZ servers.

Confirmed IoCs included repeated failed logins from specific IPs, unauthorized privilege changes, and unusual scanning patterns on DMZ servers.

7. RECOMMENDATION

To make the system safer, start by fixing technical issues like securing SSH access, closing unnecessary ports, and improving firewall rules. Also improve processes by keeping logs for a longer time, having clear steps to follow during security problems, and making sure everyone knows who to alert. For monitoring, adjust the alert system so it only flags real threats and learn what normal activity looks like. In the long term, regularly update and patch systems, redesign the network to protect important assets, and train staff on security best practices.

8. CONCLUSION

The capstone showed that Wazuh can effectively detect suspicious activity, generate alerts, and respond to potential threats in real time. The lab mirrored real-world Security Operations Center (SOC) work by collecting logs, correlating events, and taking action on incidents, just like a professional team would. Key takeaways for the organization include the importance of monitoring systems continuously, tuning alerts to focus on real threats, keeping software up to date, and training staff to follow proper security procedures.

9. REFERENCES

<https://documentation.wazuh.com/current/user-manual/capabilities/active-response/index.html>

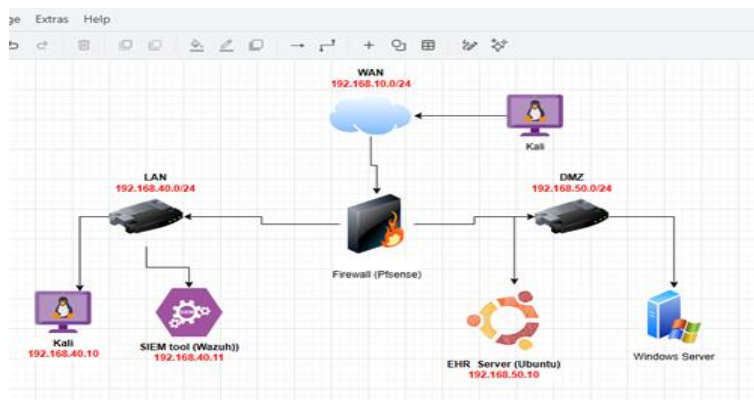
<https://docs.netgate.com/pfsense/en/latest/>

<https://documentation.help/Wireshark/>

<https://www.kali.org/tools/nmap/>

10. APPENDICES

Appendix A — Network Design & Firewall Configuration



Interfaces ⚙️ - ✕			
 WAN	↑	1000baseT <full-duplex>	192.168.10.4
 LAN	↑	1000baseT <full-duplex>	192.168.40.1
 DMZ	↑	1000baseT <full-duplex>	192.168.50.1

```

pfSense [Running] - Oracle VirtualBox
File Machine View Input Devices Help
*** Welcome to pfSense 2.8.1-RELEASE (amd64) on MedSecure ***

WAN (wan) -> em0 -> v4/DHCP4: 192.168.10.4/24
LAN (lan) -> em1 -> v4: 192.168.40.1/24
DMZ (opt1) -> em2 -> v4: 192.168.50.1/24

0) Logout / Disconnect SSH          9) pfTop
1) Assign Interfaces                10) Filter Logs
2) Set interface(s) IP address      11) Restart GUI
3) Reset admin account and password 12) PHP shell + pfSense tools
4) Reset to factory defaults        13) Update from console
5) Reboot system                    14) Enable Secure Shell (sshd)
6) Halt system                      15) Restore recent configuration
7) Ping host                        16) Restart PHP-FPM
8) Shell

Enter an option:
Message from syslogd@MedSecure at Dec 10 00:14:21 ...
php-fpm[435]: /index.php: Successful login for user 'admin' from: 192.168.40.11
(Local Database)

Message from syslogd@MedSecure at Dec 10 01:26:32 ...
php-fpm[434]: /interfaces_assign.php: Successful login for user 'admin' from: 192.168.40.11 (Local Database)

```

Firewall / Rules / LAN 🔍 📄 ?

Floating WAN LAN DMZ

Rules (Drag to Change Order)

☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☒	✓ 0/84 KIB	*	*	*	LAN Address	443	*	*		Anti-Lockout Rule	⚙️
☐	✗ 0/504 B	IPv4 *	LAN subnets	*	DMZ subnets	*	*	none			🔗 ⚙️ 📄 🗑️ ✕
☐	✓ 5/368.18 MiB	IPv4 *	LAN subnets	*	*	*	*	none		Default allow LAN to any rule	🔗 ⚙️ 📄 🗑️ ✕
☐	✓ 0/0 B	IPv6 *	LAN subnets	*	*	*	*	none		Default allow LAN IPv6 to any rule	🔗 ⚙️ 📄 🗑️ ✕

⬆️ Add
⬆️ Add
🗑️ Delete
🔄 Toggle
📄 Copy
💾 Save
⬆️ Separator

Firewall / Rules / DMZ

FloatingWANLANDMZ

Rules (Drag to Change Order)

	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☐	✓	2/285.50 MIB	IPv4 *	DMZ subnets	*	*	*	*	none	Allow DMZ to internet	    

↑ Add

↓ Add

 Delete

 Toggle

 Copy

 Save

 Separator

Appendix B — Attack Simulation & Traffic Analysis

```
Session Actions Edit View Help
(kali@kali)-[~]
$ nano users.txt
(kali@kali)-[~]
$ cat users.txt
admin
comfort
10alytics
medsecure
nacohort
eucohort
cybersecurity
packet
bruteforce
ubuntu

(kali@kali)-[~]
$ nano password.txt
(kali@kali)-[~]
$ cat password.txt
password12
10alytics
password
ephraim
saalem
mia
canada
unitedkingdon

(kali@kali)-[~]
$ hydra -i users.txt -P password.txt -t4 -v ssh://192.168.50.10
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this
```

```
(kali@kali)-[~]
$ hydra -i users.txt -P password.txt -t4 -v ssh://192.168.50.10
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this
is non-binding, these ** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2025-12-09 10:54:36
[DATA] max 4 tasks per 1 server, overall 4 tasks, 132 login tries (l:11/p:12), ~33 tries per task
[DATA] attacking ssh://192.168.50.10:22/
[VERBOSE] Resolving addresses ... [VERBOSE] resolving done
[INFO] Testing if password authentication is supported by ssh://medsecureadmin@192.168.50.10:22
[INFO] Successful, password authentication is supported by ssh://192.168.50.10:22
[22][ssh] host: 192.168.50.10 login: medsecureadmin password: admin
[22][ssh] host: 192.168.50.10 login: medsecure password: medsecure
[STATUS] 88.00 tries/min, 88 tries in 00:01h, 44 to do in 00:01h, 4 active
[22][ssh] host: 192.168.50.10 login: ubuntu password: ubuntu
[STATUS] attack finished for 192.168.50.10 (waiting for children to complete tests)
1 of 1 target successfully completed, 3 valid passwords found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2025-12-09 10:56:06
```

```

(kali@kali)-[~]
$ sudo systemctl start ssh
[sudo] password for kali:

(kali@kali)-[~]
$ sudo systemctl status ssh
● ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; disabled; preset: disabled)
   Active: active (running) since Tue 2025-12-09 08:53:11 EST; 8s ago
  Invocation: d6267e43a7f1489081fa14264cff4201
     Docs: man:sshd(8)
           man:sshd_config(5)
   Process: 32279 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
    Main PID: 32290 (sshd)
      Tasks: 1 (limit: 2118)
     Memory: 5.1M (peak: 5.6M)
        CPU: 52ms
    CGroup: /system.slice/ssh.service
            └─32290 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Dec 09 08:53:10 kali systemd[1]: Starting ssh.service - OpenBSD Secure Shell server ...
Dec 09 08:53:11 kali sshd[32290]: Server listening on 0.0.0.0 port 22.
Dec 09 08:53:11 kali sshd[32290]: Server listening on :: port 22.
Dec 09 08:53:11 kali systemd[1]: Started ssh.service - OpenBSD Secure Shell server.

(kali@kali)-[~]
$ nmap -help
Nmap 7.95 ( https://nmap.org )
Usage: nmap [Scan Type(s)] [Options] [target specification]
TARGET SPECIFICATION:
  Can pass hostnames, IP addresses, networks, etc.
  Ex: scanme.nmap.org, microsoft.com/24, 192.168.0.1; 10.0.0-255.1-254
  -iL <inputfilename>: Input from list of hosts/networks
  -iR <num hosts>: Choose random targets
  --exclude <host1[,host2][,host3], ...>: Exclude hosts/networks

```

```

(kali@kali)-[~]
$ nmap 192.168.40.0/24
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-09 09:09 EST
Nmap scan report for MedSecure.home.arpa (192.168.40.1)
Host is up (0.00096s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
53/tcp    open  domain
80/tcp    open  http
443/tcp   open  https
MAC Address: 08:00:27:39:EE:70 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap scan report for 192.168.40.10
Host is up (0.00059s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh
443/tcp   open  https
MAC Address: 08:00:27:32:25:68 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap scan report for 192.168.40.11
Host is up (0.00012s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
22/tcp    open  ssh

Nmap done: 256 IP addresses (3 hosts up) scanned in 7.72 seconds

(kali@kali)-[~]
$ nmap -sS 192.168.40.1
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-09 09:13 EST
Nmap scan report for MedSecure.home.arpa (192.168.40.1)
Host is up (0.00047s latency).

```

```

Nmap scan report for MedSecure.home.arpa (192.168.40.1)
Host is up (0.00047s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
53/tcp    open  domain
80/tcp    open  http
443/tcp   open  https
MAC Address: 08:00:27:39:EE:70 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 5.35 seconds

(kali@kali)-[~]
$ nmap -sV 192.168.40.1
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-09 09:14 EST
Nmap scan report for MedSecure.home.arpa (192.168.40.1)
Host is up (0.00057s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE  VERSION
53/tcp    open  domain   Unbound
80/tcp    open  http     nginx
443/tcp   open  ssl/http nginx
MAC Address: 08:00:27:39:EE:70 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 17.95 seconds

(kali@kali)-[~]
$ nmap -sU 192.168.40.1
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-09 09:14 EST
Nmap scan report for MedSecure.home.arpa (192.168.40.1)
Host is up (0.00055s latency).
Not shown: 997 open|filtered udp ports (no-response)
PORT      STATE SERVICE
53/udp    open  domain
123/udp   open  ntp

```

```

(kali@kali)-[~]
$ nmap -sU 192.168.40.1
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-09 09:14 EST
Nmap scan report for MedSecure.home.arpa (192.168.40.1)
Host is up (0.00055s latency).
Not shown: 997 open|filtered udp ports (no-response)
PORT      STATE SERVICE
53/udp    open  domain
123/udp   open  ntp
161/udp   open  snmp
MAC Address: 08:00:27:39:EE:70 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 13.73 seconds

(kali@kali)-[~]
$ nmap -sU -p 161 --script=snmp.interfaces 192.168.40.1
Starting Nmap 7.95 ( https://nmap.org ) at 2025-12-09 09:26 EST
Nmap scan report for MedSecure.home.arpa (192.168.40.1)
Host is up (0.00050s latency).

PORT      STATE SERVICE
161/udp   open  snmp
snmp.interfaces:
  em0
    IP address: 192.168.10.4 Netmask: 255.255.255.0
    MAC address: 08:00:27:87:cc:96 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
    Type: ethernetCsmacd Speed: 1 Gbps
    Status: up
    Traffic stats: 899.64 Kb sent, 9.64 Mb received
  em1
    IP address: 192.168.40.1 Netmask: 255.255.255.0
    MAC address: 08:00:27:39:ee:70 (PCS Systemtechnik/Oracle VirtualBox virtual NIC)
    Type: ethernetCsmacd Speed: 1 Gbps

```

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	192.168.50.10	192.168.40.11	SSH	118	Server:
2	0.000196838	192.168.40.11	192.168.50.10	SSH	118	Client:
3	0.000223610	192.168.40.11	192.168.50.10	TCP	66	38850 →
4	0.000484363	192.168.40.11	192.168.50.10	TCP	74	49122 →
5	0.002691847	192.168.50.10	192.168.40.11	TCP	66	22 → 38
6	0.002692163	192.168.50.10	192.168.40.11	TCP	74	22 → 49
7	0.002744847	192.168.40.11	192.168.50.10	TCP	66	49122 →
8	0.002800773	192.168.40.11	192.168.50.10	SSHv2	89	Client:
9	0.006356746	192.168.50.10	192.168.40.11	TCP	66	22 → 38
10	0.006357019	192.168.50.10	192.168.40.11	TCP	66	22 → 49
11	0.006394322	192.168.40.11	192.168.50.10	TCP	66	38850 →

Frame 2: 118 bytes on wire (944 bits), 118 bytes captured (944 bits) on interface 0
 Ethernet II, Src: PCSSystemtec_1f:b7:23 (08:00:00:00:00:00), Dst: 192.168.40.11 (08:00:00:00:00:00)
 Internet Protocol Version 4, Src: 192.168.40.11, Dst: 192.168.50.10
 Transmission Control Protocol, Src Port: 38850, Dst Port: 22
 SSH Protocol

0000 08 00 27 39 ee 70 08 00 27 1f b7 23 00 00 00
 0010 00 68 4c 01 40 00 40 06 13 29 c0 a8 20 00 00
 0020 32 0a 97 c2 00 16 80 0a d4 81 f4 75 30 00 00
 0030 01 02 db c0 00 00 01 01 08 0a aa a8 d4 81 f4
 0040 04 e6 94 91 35 29 fb 97 9d fe 3b 6c 20 00 00
 0050 a2 48 e4 20 f8 49 9b de 3b d0 b3 d4 b3 d0 b3
 0060 6c 8b bf b4 40 2c 3f be 02 f4 0f 92 f4 0f 92
 0070 98 eb d7 dd f4 ff

wireshark_eth004Q1G3.pcapng | Packets: 24196 · Dropped: 0 (0.0%) | Profile: Default

PCAP Analysis Snippet

The packet capture showed repeated SSH login attempts coming from a single source, which is consistent with a brute-force attack. The traffic included multiple failed authentication packets sent in rapid succession, indicating that an automated tool was trying different username and password combinations. No successful login was observed in the capture, but the volume and pattern of attempts clearly show malicious behavior rather than normal user activity.

Additionally, there were signs of port-scanning activity, where the attacker probed several open ports on the server to identify potential weaknesses. Overall, the pcap confirms that the server in the DMZ was being targeted by repeated unauthorized access attempts, supporting the alerts detected by Wazuh during the monitoring phase.

Appendix C — Wazuh SIEM Detection & Response

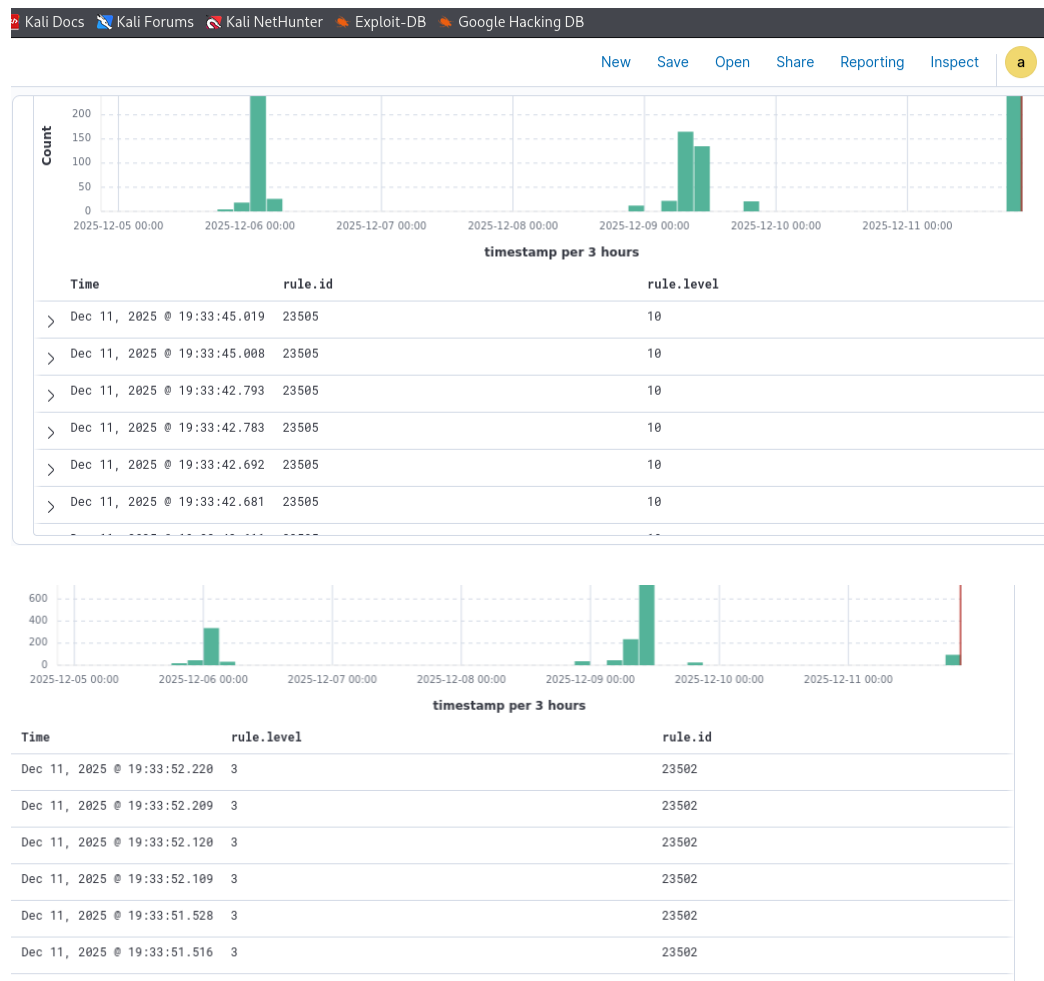
```
Valid_HTT forever preferred_HTT forever
[root@wazuh-server ~]# [12648.916468] clocksource: Long readout interval, skipping watchdog check:
cs_nsec: 11396331524 wd_nsec: 11396323213
[14235.989575] 23:30:49.450440 timesync vgsvcTimeSyncWorker: Radical host time change: 22 924 486 00
0 000ns (HostNow=1 765 323 043 177 000 000 ns HostLast=1 765 300 118 691 000 000 ns)
[14245.996372] 23:30:59.452776 timesync vgsvcTimeSyncWorker: Radical guest time change: 23 853 054 3
07 000ns (GuestNow=1 765 323 059 452 763 000 ns GuestLast=1 765 299 206 398 456 000 ns fSetTimeLastL
oop=true)

[root@wazuh-server ~]# systemctl restart wazuh-manager
[root@wazuh-server ~]# [22749.722758] 01:52:42.487338 control Session 0 is about to close ...
[22749.745934] 01:52:42.546078 control Stopping all guest processes ...
[22749.752829] 01:52:42.550189 control Closing all guest files ...
[22749.761577] 01:52:42.561880 control vbgIR3GuestCtrlDetectPeekGetCancelSupport: Supported (#1)
[22749.953385] e1000: eth0 NIC Link is Down
[22750.387017] 00:00:17.962514 timesync vgsvcTimeSyncWorker: Radical host time change: 166 064 093 0
00 000ns (HostNow=1 765 497 617 962 000 000 ns HostLast=1 765 331 553 869 000 000 ns)
[22756.184863] e1000: eth0 NIC Link is Up 1000 Mbps Full Duplex, Flow Control: RX
[22760.475366] 00:00:28.047158 timesync vgsvcTimeSyncWorker: Radical guest time change: 166 064 861
539 000ns (GuestNow=1 765 497 628 047 141 000 ns GuestLast=1 765 331 563 185 602 000 ns fSetTimeLastL
oop=true)
```

< Manager configuration

Edit ossec.conf of Manager

```
228 <command>
229   <name>netsh</name>
230   <executable>netsh.exe</executable>
231   <timeout_allowed>yes</timeout_allowed>
232 </command>
233
234 <active-response>
235   <disabled>no</disabled>
236   <command>firewall-drop</command>
237   <location>local</location>
238   <rules_group>sshd</rules_group>
239   <timeout>0</timeout>
240 </active-response>
```

The alerts from Wazuh were matched with actual network activity to see what was really happening. For example, alerts about failed logins lined up with network logs showing one IP trying to connect many times, which looked like a brute-force attack. Alerts about scanning matched unusual port checks on the DMZ server, and privilege escalation alerts matched system logs showing users trying to gain higher access. By connecting the alerts to real evidence, we could see which events were real threats and decide how to respond.

11. SELF-REFLECTION

- What was easy? – Drawing the network and installing the applications and setting up LAN, WAN and DMZ were quite easy for me
- What was challenging? – For me, the challenging part was copy and paste on Ubuntu and interpreting pcap files.

- What did you learn about attackers? – During this project I learnt that attackers often start by gathering information, such as scanning the network to see which systems and ports are open. They tend to repeat the same actions many times, like trying multiple logins attempts in a short period, showing persistence in finding weak points. The tests also showed that attackers commonly take advantage of simple misconfigurations, such as weak passwords, unnecessary open ports, or overly permissive access rules.
- What did you learn about defenders / SOC operations? - I learnt that security teams need good visibility into what's happening on the network at all times. They rely on alerts and logs to notice when something looks unusual. Defenders also need to adjust their tools so they don't get overwhelmed with too many unimportant alerts. Quick action is important, because the longer a threat goes unnoticed, the more damage it can cause. Overall, I learnt that defenders must stay alert, communicate well, and keep improving their tools and processes to stay ahead of attackers.
- How would you improve your lab setup or methodology for future exercises?